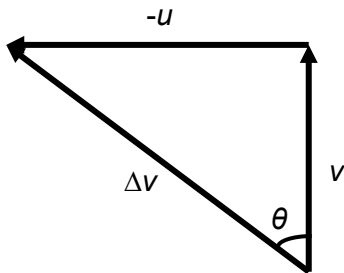


- 2 D** Initial velocity, $u = 4.0 \text{ m s}^{-1}$ due east.
Final velocity, $v = 3.0 \text{ m s}^{-1}$ due north.
Change in velocity $\Delta v = v - u = v + (-u)$, where $(-u) = 4.0 \text{ m s}^{-1}$ due west.
Doing a vector addition:



$$|\Delta v| = \sqrt{(-u)^2 + v^2} = \sqrt{4^2 + 3^2} = 5$$

$$\theta = \tan^{-1}(4/3) = 53^\circ$$

Hence, the change in velocity is 53° west of north.